

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Annexure A

18
20
Total Marks:30

Constraint-Based Problem Solving

Code of Conduct:

I V. Sree Ruthin (192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sree Ruthin

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Rural Branch office - Internet Connections, Devices and Security

The company is located in a rural area where high-speed wired internet is unavailable. The network must support cloud applications, video conferencing and secure communication with the headquarters.

A wireless router can provide internet connectivity, while a managed switch can connect office computers and other wired devices. A wifi access point can provide wireless access to employees. A VPN can create a secure connection between the branch office and headquarters.

For security, a firewall should be configured to block unauthorized traffic. WPA3 encryption can be used for wifi, while strong passwords and user authentication should be implemented. Regular software and firmware updates should also be applied.

The main limitation is bandwidth, therefore video conferencing should be given higher priority using QoS. Network monitoring can help identify heavy traffic. As the company grows, additional bandwidth, access points or a second internet connection can be added.

Scalability upgrade bandwidth and add devices when required.

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Communication - Secure Network Architecture
The financial institution handles sensitive customer information, so security must be the main priority without significantly reducing network performance or increasing operations costs. The network should be divided into separate security zones using VLANs.

A suitable architecture would include separate VLANs for employees, servers, guests users and administrative systems. Sensitive banking servers should be placed in a DMZ or protected server network. This prevents direct access from normal user networks.

A firewall should be placed at the internet boundary to inspect incoming and outgoing traffic. Additional internal firewall rules can control communication between VLAN. Only necessary services should be allowed.

Employees should use strong authentication. Preferably to inspect incoming and outgoing traffic should be blocked and network monitoring and be used.

An IDS can monitor network traffic and identify suspicious activity. An IPS can additionally block detected threats.

To maintain performance, firewall and security rules should be optimized, unnecessary traffic should be blocked and network monitoring should be used to identify bottlenecks.

Recommended architecture

Internet → Firewall → Core Switch → VLANs.

VLAN 1: Employees

VLAN 2: Banking / Application servers.

VLAN 3: Guest Network

VLAN 4: Administration

Security: Firewall + IDS/IPS

Authentication: MFA + Strong password.